well-typed syntax
with equations

normalizálás

well-typed syntax tree

típuskikövetkeztetés

típuskitörlés

abstract binding tree

változók
elnevezése

scope
check

abstract syntax tree

parser

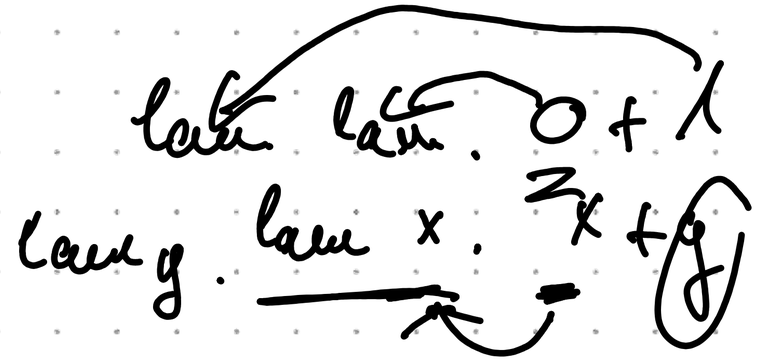
zárójelek
hozzáadása

lexikális elemek
sorozata

lexer
lexical analyzer

szóköz
hozzáadás

string



isZero$_{nat}$(num$_{nat}$ 0) -> [isZero, (, num, 0,)]

isZero$_{?}$(num, 0)

num 3 $\overset{?}{=}$ "num 3" ✓

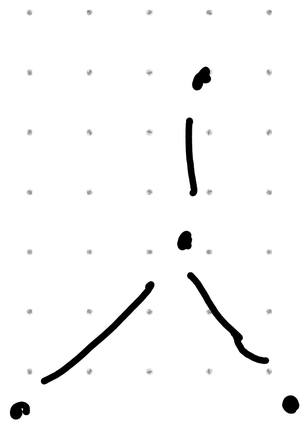
then then if $\overset{?}{=}$ "then then if" ✓

lexer

[then, then, if]

$T := \text{true} \mid \text{false} \mid \text{if } T \text{ then } T \text{ else } T \mid \text{num } \mathbb{N} \mid \text{iszero } T$

$\mid T + T$



$\not\vdash$ iszero (num 0 + num 1) : Bool

else (true × ☐ true) : Bool

$Ty := \text{Bool} \mid \text{Nat}$

. true : Tm Bool

. false : Tm Bool

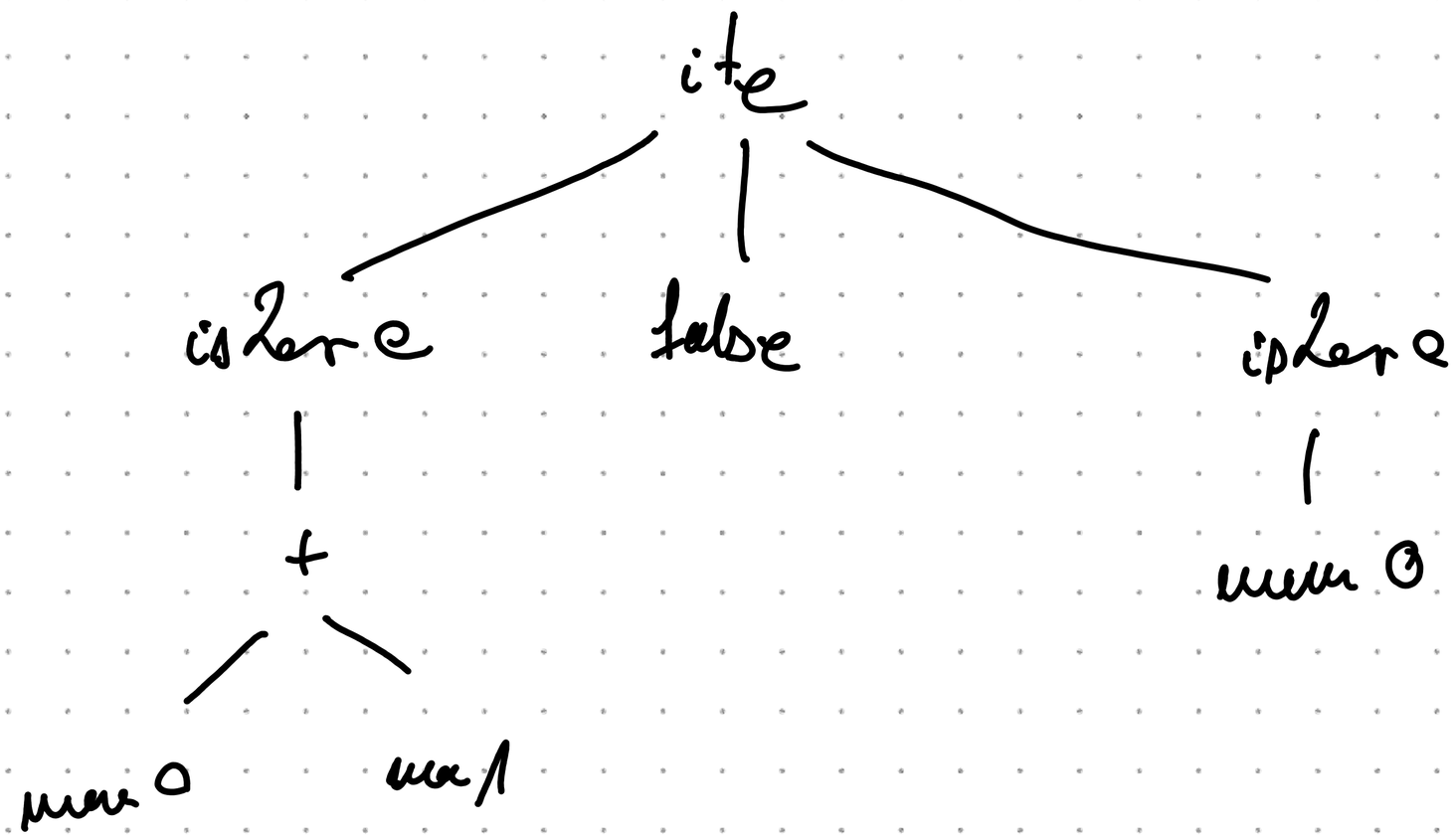
. if : $\forall A.\ \text{Tm}\ \text{Bool} \to \text{Tm}\ A \to \text{Tm}\ A \to \text{Tm}\ A$

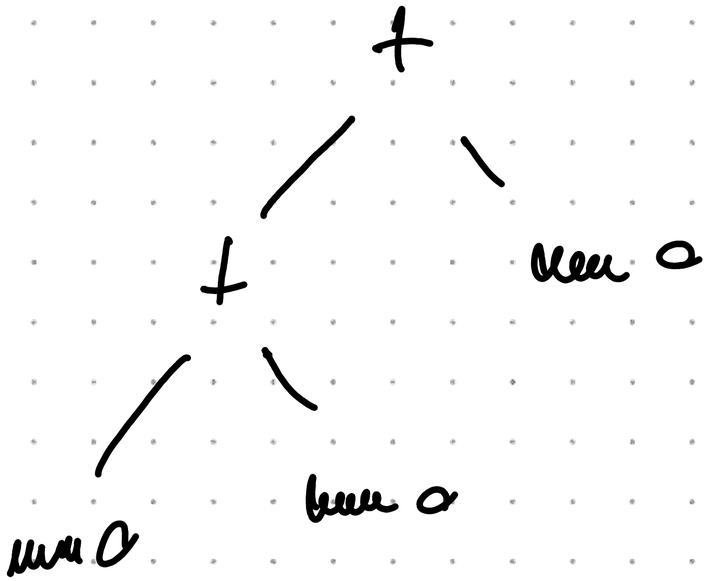
. num : $\mathbb{N} \to \text{Tm}$ Nat

. iszero : $\text{Tm}\ \text{Nat} \to \text{Tm}$ Bool

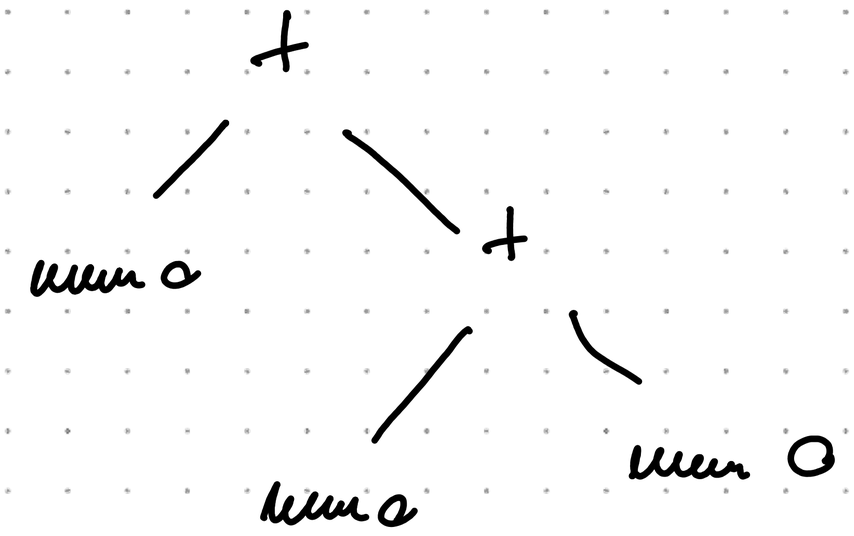
. _+_ : $\text{Tm}\ \text{Nat} \to \text{Tm}\ \text{Nat} \to \text{Tm}\ \text{Nat}$

if isZero(num 0 + num 1) then false
else isZero(num 0)





$\neq$



$(mC + mO) + nO$

$mO + (nO + nO)$

$$\mathrm{ite} : \mathrm{Tm}\ \mathrm{Bool} \to \mathrm{Tm}\ A \to \mathrm{Tm}\ A \to \mathrm{Tm}\ A$$

$$\frac{t : \mathrm{Tm}\ \mathrm{Bool} \qquad u : \mathrm{Tm}\ A \qquad v : \mathrm{Tm}\ A}{\mathrm{ite}\ t\ u\ v : \mathrm{Tm}\ A}$$

$$\frac{\checkmark}{\mathrm{true} : \mathrm{Tm}\ \mathrm{Bool}} \qquad \frac{Ax.}{\mathrm{false} : \mathrm{Tm}\ \mathrm{Bool}}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{\mathrm{succ}\ 0 : \mathrm{Tm\ Nat}} \qquad \mathrm{true} : \mathrm{Tm\ Nat}}{\mathrm{succ}\ 0 + \mathrm{true} : \mathrm{Tm\ Nat}}}{\mathrm{iszero}\ (\mathrm{succ}\ 0 + \mathrm{true}) : \mathrm{Tm\ Bool}}, \qquad \dfrac{\dfrac{}{n0 : \mathrm{Tm\ Nat}} \qquad \dfrac{}{n1 : \mathrm{Tm\ Nat}}}{n0 + n1 : \mathrm{Tm\ Nat}} \qquad \dfrac{}{\mathrm{false} : \mathrm{Tm\ Bool}} \qquad \dfrac{\dfrac{}{\mathrm{succ}\ 0 : \mathrm{Tm\ Nat}}}{\mathrm{iszero}\ (\mathrm{succ}\ 0) : \mathrm{Tm\ Bool}}}{\mathrm{ite}\ (\mathrm{iszero}\ (\mathrm{succ}\ 0 + n1))\ \mathrm{false}\ (\mathrm{iszero}\ (\mathrm{succ}\ 0)) : \mathrm{Tm\ Bool}}$$

$iteB_1$: if true then u else $v \equiv u$

$iteB_2$: if false then u else $v \equiv v$

$isZeroB_1$: $isZero\ (num\ 0) \equiv true$

$isZeroB_2$: $\forall n.\ isZero\ (num\ (S\ n)) \equiv false$

$+B$: $num\ n + num\ m \equiv num\ (n + m)$

$\equiv$ Razor +

$\equiv$ meta-level +

-> ite (isZero (succ c + succ 1)) then false

+B else isZero (succ 0)

isZero B1

|||

ite (isZero (succ 1)) then false else true

||| isZero B2

ite false then false else true

||| iteB2

true -> normal form

||| isZero B1

isZero (succ 0)

||| tB-1

isZero (succ 0 + succ 0)

->